

ARTIFICIAL INTELLIGENCE IN AN AUTONOMOUS FIRE FIGHTING ROBOT

ABSTRACT

The Main Objective of this Paper is to develop a automated system which detects and extinguishes fire using Image Processing. A fire fighter must be able to stop fire quickly and safe extinguish the fire, preventing further damage. As the technology has finally bridged the gap between fire fighting and machines allowing for a more efficient and effective method of fire fighting. We create an automated system for early detection and extinguished of fire thereby making the work of fireman easy and reduce loss of human life. The paper is designed to develop a fire fighting robot using Arduino UNO. The robotic vehicle is loaded with water pump which is controlled by servos. An ATmega 328 microcontroller is used for the desired operation. At the transmitting end using commands are sent to the receiver to control the movement of the robot either to move forward, and left or right etc. At the receiving end tow motors are interfaced to the microcontroller where two of them are used for the movement of the vehicle and the one to position the robot. The ultrasonic sensor adequate range with obstacle detection, while the receiver driver module used to drive DC motors via motor driver IC for necessary work. A water tank along with water pump is mounted on the robot body and its operation is carried out from the microcontroller output through appropriate command from the transmitting end. The whole operation is controlled by an ATmega 328 microcontroller. A motor driver IC is interfaced to the microcontroller through which the controller drives the motors, three IR flame sensors are fixed on robot chassis to sense the fire and to reach the destination to put off the fire..The automated fire-fighting robot is capable of detecting fire and extinguished the fire source successfully. The system works on real time, as it provides continuous monitoring. It will be very interesting to develop and have a useful independent fire-fighting service.

Keywords: Image Processing, Arduino UNO, USB Camera, Motor Driver, Pump.

I. INTRODUCTION

Now-a-days, Fire accidents are becoming more and more in society that maybe building fires, motor vehicle fires, etc. These accidents cause many hazards to people and also pollute and damage to environment. Fire accident leads to number of people deaths. Basically, the firefighter's works hard to protect the people from the fire and to save human and animals life. But they cannot protect the more number of people from the fire and it also somewhat time taken to fire fighters. So we came with application of firefighting robot based on artificial intelligence.. As the technology has finally bridged the gap between fire fighting and machines allowing for a more efficient and effective method of fire fighting [1]. We create an automated system for early detection and extinguished of fire thereby making the work of fireman easy and reduce loss of human life. The importance of the proposed thesis is to make a reliable, safe, and smart system to reduce limitations and faults like false alarms, which cause panic among the people and even the loss of money with the use of new technology and make the places safe from the hazardous fire. Additionally, having a compact size and automatic small and narrow spaces with hazardous environments such as tunnels or nuclear power plants [6]. Furthermore, this robot can increase productivity, safe, efficiency and quality of the task given [2]. A human operator can monitor the robot by using camera which connects to a smart phone or remote devices .Finally, robots can be used in situations where human access is not possible, such as in collapsed buildings or in remote locations. The paper aims to motivate the robotics community to develop a real-world application based on what it can accomplish.

In addition to being able to be installed in homes, laboratories, stores, shops etc..., fire fighting robot is easily portable and can be used once installed [3].

II. FUTURE SCOPE

The paper has been motivated by the desire to design a system that can detect fires and take appropriate action, without any human intervention. The development of sensor networks and the maturity of robotics suggest that we can use mobile agents for tasks that involve perception of an external stimulus and reacting to the stimulus, even when the reaction involves a significant amount of mechanical actions[4]. This provides us the opportunity to pass on to robots tasks that traditionally humans had to do but were inherently life-threatening. Fire-fighting is an obvious candidate for such automation. Given the number of lives lost regularly in fire-fighting, the system we envision is crying for adoption. Our experience suggests that designing a fire-fighting system with sensors and robots is within the reach of the current sensor network and mobile agent technologies. Furthermore, we believe that the techniques developed in this work will carry over to other areas involving sensing and reacting to stimulus, where we desire to replace the human with an automated mobile agent[5].

III. REQUIREMENTS



Figure 1: Arduino UNO

It was also the first USB board released by Arduino. It is considered as the powerful board used in various papers [6]. Arduino.cc developed the Arduino UNO board. Arduino UNO is based on an Atmega328P microcontroller. It is easy to use compared to other boards, such as the Arduino Mega board, etc. The board consists of digital and analog Input/Output pins (I/O), shields, and other circuits[6]. The Arduino UNO includes 6 analog pin inputs, 14 digital pins, a USB connector, a power jack, and an ICSP (In-Circuit Serial Programming) header. It is programmed based on IDE, which stands for Integrated Development Environment. It can run on both online and offline platforms[7].



Figure 2: ESP32-CAM

ESP32-CAM is a low-cost ESP32-based development board with onboard camera, small in size. It is an ideal solution for IoT application, prototypes constructions and DIY papers. The board integrates WiFi, traditional Bluetooth and low power BLE, with 2 high-performance 32-bit LX6 CPUs. It adopts 7-stage pipeline

architecture, on-chip sensor, Hall sensor, temperature sensor and so on, and its main frequency adjustment ranges from 80MHz to 240MHz[8].

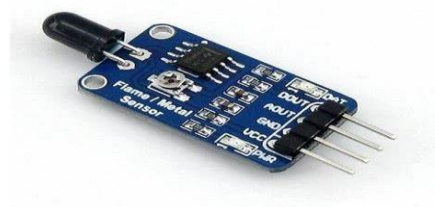
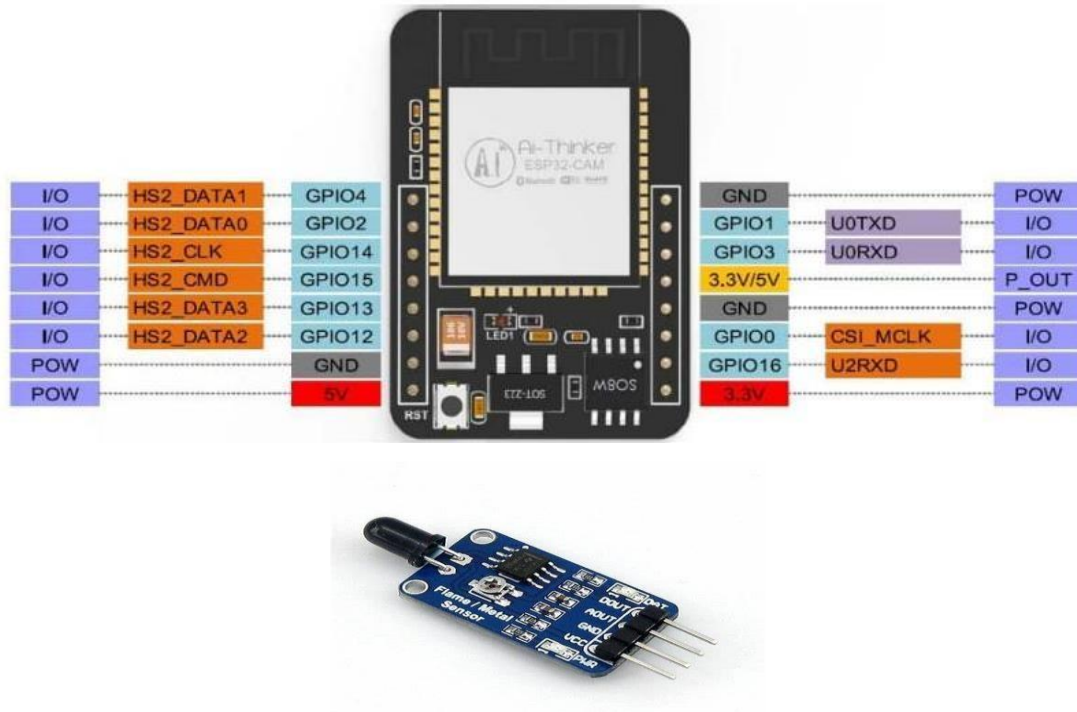


Figure 3: Fire Sensor

Fire or flame sensor module can detect flames in the 760-1100 nanometre wavelength range[9]. Small flames like a lighter flame can be detected at roughly 0.8m. The detection angle is roughly 60 degrees and the sensor is particularly sensitive to the flame spectrum. The sensor can be a great addition in a fire fighting robot ,it can be used as a robot eyes to find the fire source. When the sensor detects flame the signal LED will light up and D0 pin goes low.



Figure 4: DC Motor

The basic principle of DC motor is that whenever a current carrying conductor places in the magnetic field, it experiences a mechanical force. The main benefit of using DC motors in robotics is their high torques [2]. It converts direct current DC electrical energy into mechanical energy [10].



Figure 5: Pump

The water pump is operated at 5v which can be interfaced with Arduino.



Figure 6: Servo motor

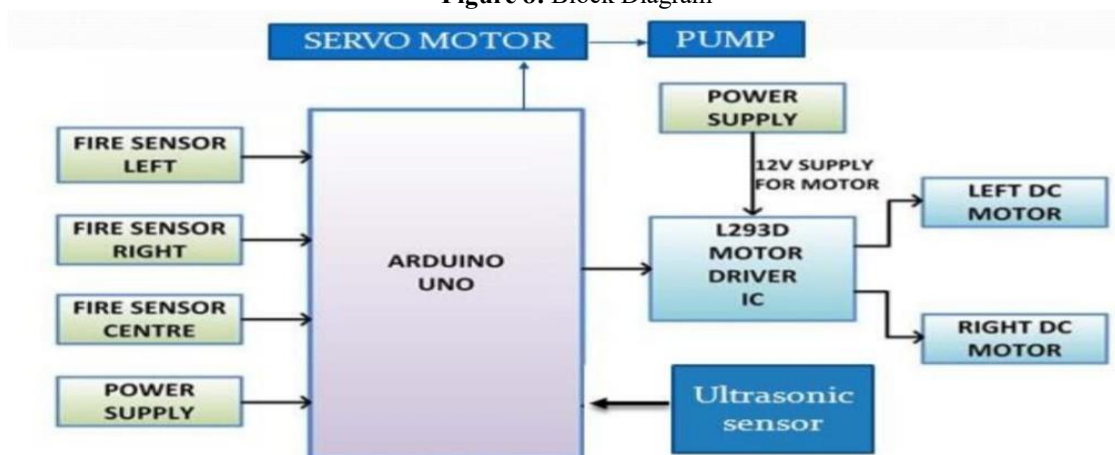
A servo is a small DC motor with the following components added: some gear reduction, a position sensor on the motor shaft, and an electronic circuit that controls the motor's operation. In other words, a servo is to a DC motor what the Arduino is the Atmega microcontroller---components and housing that make the motor easy to use[11].



Figure 7: Thermocouple

We are using 1000°C Thermocouple and 100°C Thermocouple using that 1000°C Thermocouple outside the robot we can able to gather the outside temperature of the robot in which the robot is working and by using the 100°C Thermocouple we can able to gather the interior temperature of the robot the component which have been located inside the robot by knowing the current temperature of the robot we can able to operate the robot in most safe manner.

Figure 8: Block Diagram



IV. WORKING

The robotic vehicle is loaded with water pump which is controlled by servos. An ATmega 328 microcontroller is used for the desired operation. At the transmitting end using commands are sent to the receiver to control the movement of the robot either to move forward, and left or right etc. At the receiving end two motors are

interfaced to the microcontroller where two of them are used for the movement of the vehicle and the one to position the robot. The ultrasonic sensor adequate range with obstacle detection, while the receiver driver module used to drive DC motors via motor driver IC for necessary work[12]. A water tank along with water pump is mounted on the robot body and its operation is carried out from the microcontroller output through appropriate command from the transmitting end. The whole operation is controlled by an Atmega 328 microcontroller. A motor driver IC is interfaced to the microcontroller through which the controller drives the motors, three ir flame sensors are fixed on robot chassis to senses the fire and to reach the destination to putoff the fire. The automated fire-fighting robot is capable of detecting fire and extinguished the fire source successfully[13].

Figure 9: Circuit Diagram

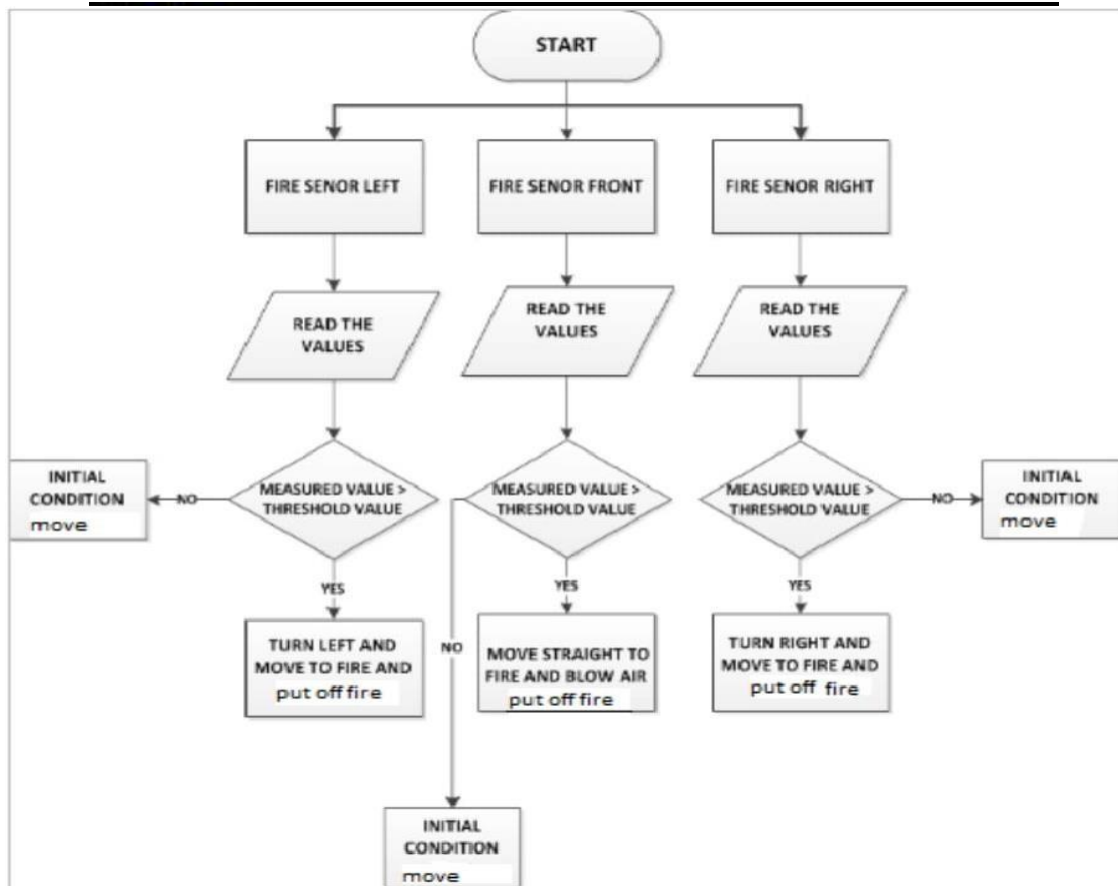


Figure 10: Flow Chart

V. EXPERIMENTAL RESULTS



Figure 11: Prototype of Autonomous Fire Fighting Robot

VI. CONCLUSION

The automated fire fighting robot is capable of detecting fire and extinguishes the fire source successfully. The prototype of the fire fighter robot was efficiently designed. This prototype has facilities to be integrated with many sensors making it move forward. The toolkit detects the infrared light emitted by the fire with photo diode and sends signal to controller. The system works on real time, as it provides continuous monitoring. For detecting the fire in the peripheral region of robot, fire sensors are used, which helps in efficient working of the robot and to avoid damage to the robot.

VII. REFERENCES

- [1] Jeelani, S., et al., Robotics and medicine: A scientific rainbow in hospital. Journal of Pharmacy & Bioallied Sciences, 2015. 7(Suppl 2): p.S381-S383.
 - [2] Bugatha Ram Vara prasad, "Solar Powered BLDC Motor with HCC Fed Water Pumping System for Irrigation," Int. J. Res. Appl. Sci. Eng. Technol., vol. 7, no. 3, pp. 788–796, 2019, doi: 10.22214/ijraset.2019.3137.
 - [3] Aliff, M., S. Dohta, and T. Akagi, Simple Trajectory Control Method of Robot Arm Using Flexible Pneumatic Cylinders. Journal of Robotics and Mechatronics, 2015. 27(6): p. 698-705.
 - [4] Aliff M, D.S., and Akagi T, Control and analysis of simple-structured robot arm using flexible pneumatic cylinders. International Journal of Advanced and Applied Sciences, 2017. 4(12): p. 151-157.
 - [5] Aliff, M., S. Dohta, and T. Akagi, Control and analysis of robot arm using flexible pneumatic cylinder. Mechanical Engineering Journal, 2014. 1(5): p. DR0051-DR0051.
 - [6] Bugatha Ram Vara prasad, "HIGHWAY MONITORING SYSTEM AND POWER SAVING," vol. 8, no. 4, pp. 2270–2274, 2020.
 - [7] M. Aliff, S. Dohta and T. Akagi, Trajectory controls and its analysis for robot arm using flexible pneumatic cylinders," IEEE International Symposium on Robotics and Intelligent Sensors (IRIS), 2015, pp. 48-54.
 - [8] M. Aliff, S. Dohta and T. Akagi, Trajectory control of robot arm using flexible pneumatic cylinders and embedded controller, IEEE International Conference on Advanced Intelligent Mechatronics (AIM), 2015, pp. 1120-1125.
 - [9] C. Xin, D. Qiao, S. Hongjie, L. Chunhe and Z. Haikuan, Design and Implementation of Debris Search and Rescue Robot System Based on Internet of Things, International Conference on Smart Grid and Electrical Automation (ICSGEA), 2018, pp. 303-307.
-

- [10] Bugatha Ram Vara prasad, D. V. S. J. Poojitha, and K. Suneetha, "Closed-Loop Control of BLDC Motor Driven Solar PV Array Using Zeta Converter Fed Water Pumping System," vol. 04, no. 17, pp. 2795–2803, 2017.
 - [11] Yusof, M., and Dodd, T., Pangolin: A Variable Geometry Tracked Vehicle With Independent Track Control, *Field Robotics*, pp. 917-924.
 - [12] Day, C.-P., *Robotics in Industry—Their Role in Intelligent Manufacturing*. Engineering, 2018. 4(4): p. 440-445.
 - [13] J. Lee, G. Park, J. Shin and J. Woo, Industrial robot calibration method using denavit — Hatenberg parameters, 17th International Conference on Control, Automation and Systems (ICCAS), 2017, pp. 1834-1837.
-